Exercises 5.5

Evaluating Indefinite Integrals

Evaluate the indefinite integrals in Exercises 1–16 by using the given substitutions to reduce the integrals to standard form.

6. $\displaystyle\int \frac{(1+\sqrt{x})^{1/3}}{\sqrt{x}}\,dx, \quad u = 1+\sqrt{x}$

13. $\displaystyle\int \sqrt{x}\sin^2(x^{3/2}-1)\,dx, \quad u = x^{3/2}-1$

Evaluate the integrals in Exercises 17–50.

24. $\displaystyle\int \tan^2 x \sec^2 x\,dx$

40. $\displaystyle\int \frac{1}{x^3}\sqrt{\frac{x^2-1}{x^2}}\,dx$

Evaluate the integrals in Exercises 53 and 54.

54. $\int \frac{\sin \sqrt{\theta}}{\sqrt{\theta}\cos^3 \sqrt{\theta}}\, d\theta$

Exercises 5.6

Evaluating Definite Integrals

Use the Substitution Formula in Theorem 7 to evaluate the integrals in Exercises 1–24.

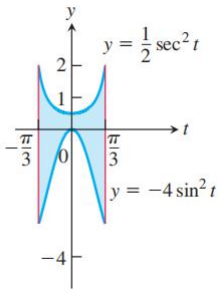
8. **b.** $\int_1^4 \frac{10\sqrt{v}}{(1 + v^{3/2})^2}\, dv$

13. a. $\int_0^{2\pi} \frac{\cos z}{\sqrt{4 + 3\sin z}}\, dz$

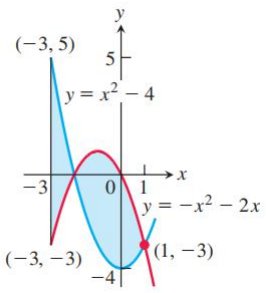
Area

Find the total areas of the shaded regions in Exercises 25–40.

30.



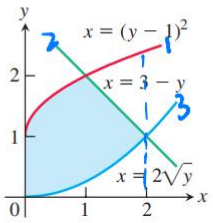
37.



Find the areas of the regions enclosed by the lines and curves in Exercises 41–50.

50. $y = |x^2 - 4|$ and $y = (x^2/2) + 4$

78. Find the area of the region in the first quadrant bounded on the left by the y-axis, below by the curve $x = 2\sqrt{y}$, above left by the curve $x = (y - 1)^2$, and above right by the line $x = 3 - y$.



84. Show that if f is continuous, then

$$\int_0^1 f(x)\,dx = \int_0^1 f(1 - x)\,dx.$$

88. By using a substitution, prove that for all positive numbers x and y,

$$\int_x^{xy} \frac{1}{t}dt = \int_1^y \frac{1}{t}dt.$$

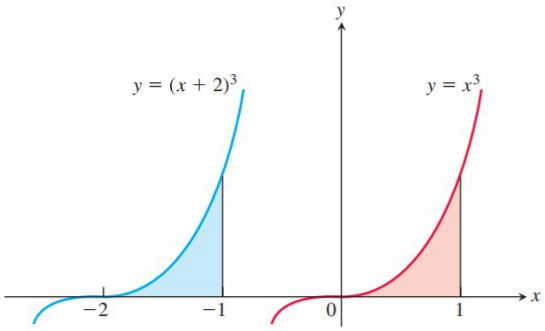
The Shift Property for Definite Integrals A basic property of definite integrals is their invariance under translation, as expressed by the equation

$$\int_a^b f(x)\,dx = \int_{a-c}^{b-c} f(x+c)\,dx. \tag{1}$$

The equation holds whenever f is integrable and defined for the necessary values of x. For example in the accompanying figure, show that

$$\int_{-2}^{-1} (x+2)^3\,dx = \int_0^1 x^3\,dx$$

because the areas of the shaded regions are congruent.



Chapter 5 Practice Exercises

Definite Integrals

In Exercises 5–8, express each limit as a definite integral. Then evaluate the integral to find the value of the limit. In each case, P is a partition of the given interval and the numbers c_k are chosen from the subintervals of P.

6. $\lim_{\|P\|\to 0} \sum_{k=1}^{n} c_k(c_k{}^2 - 1)^{1/3}\,\Delta x_k$, where P is a partition of $[1, 3]$

Find the areas of the regions enclosed by the curves and lines in Exercises 15–26.

24. $y = |\sin x|, \quad y = 1, \quad -\pi/2 \le x \le \pi/2$

30. Find the area of the region cut from the first quadrant by the curve $x^{1/2} + y^{1/2} = a^{1/2}$.

Evaluating Indefinite Integrals

Evaluate the integrals in Exercises 37–44.

44. $\int \csc\theta \cot\theta \sqrt{1 - \csc\theta}\, d\theta$

Evaluating Definite Integrals

Evaluate the integrals in Exercises 45–66.

52. $\int_0^2 \frac{dr}{\sqrt[3]{(9 + 7r)^2}}$

80. Find dy/dx if $y = \int_{\cos x}^{0} (1/(1 - t^2))\, dt$. Explain the main steps in your calculation.

Chapter 5 Additional and Advanced Exercises

3. Initial value problem Show that

$$y = \frac{1}{a}\int_0^x f(t)\sin a(x - t)\,dt$$

solves the initial value problem

$$\frac{d^2y}{dx^2} + a^2y = f(x), \qquad \frac{dy}{dx} = 0 \text{ and } y = 0 \text{ when } x = 0.$$

(*Hint:* $\sin(ax - at) = \sin ax \cos at - \cos ax \sin at$.)

6. Find $f(\pi/2)$ from the following information.

i) f is positive and continuous.

ii) The area under the curve $y = f(x)$ from $x = 0$ to $x = a$ is

$$\frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a.$$

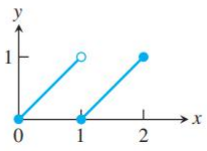
8. Prove that

$$\int_0^x \left(\int_0^u f(t)\, dt \right) du = \int_0^x f(u)(x - u)\, du.$$

(*Hint:* Express the integral on the right-hand side as the difference of two integrals. Then show that both sides of the equation have the same derivative with respect to x.)

(In Ex.8, assume f is continuous.)

17. Find the average value of the function graphed in the accompanying figure.



21. Let $f(x)$ be a continuous function. Express

$$\lim_{n\to\infty} \frac{1}{n}\left[f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \cdots + f\left(\frac{n}{n}\right)\right]$$

as a definite integral.

21. Let $f(x)$ be a continuous function. Express

$$\lim_{n\to\infty} \frac{1}{n}\left[f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \cdots + f\left(\frac{n}{n}\right)\right]$$

as a definite integral.

(不用做 Ex.21)

22. Use the result of Exercise 21 to evaluate

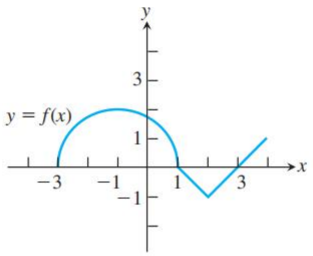
c. $\lim_{n\to\infty} \frac{1}{n}\left(\sin\frac{\pi}{n} + \sin\frac{2\pi}{n} + \sin\frac{3\pi}{n} + \cdots + \sin\frac{n\pi}{n}\right)$.

What can be said about the following limits?

d. $\lim_{n\to\infty} \frac{1}{n^{17}}(1^{15} + 2^{15} + 3^{15} + \cdots + n^{15})$

Defining Functions Using the Fundamental Theorem

25. A function defined by an integral The graph of a function f consists of a semicircle and two line segments as shown. Let $g(x) = \int_1^x f(t)\,dt$.



a. Find $g(1)$. **b.** Find $g(3)$. **c.** Find $g(-1)$.

d. Find all values of x on the open interval $(-3, 4)$ at which g has a relative maximum.

e. Write an equation for the line tangent to the graph of g at $x = -1$.

f. Find the x-coordinate of each point of inflection of the graph of g on the open interval $(-3, 4)$.

g. Find the range of g.

Leibniz's Rule In applications, we sometimes encounter functions defined by integrals that have variable upper limits of integration and variable lower limits of integration at the same time. We can find the derivative of such an integral by a formula called **Leibniz's Rule**.

Leibniz's Rule

If f is continuous on $[a, b]$ and if $u(x)$ and $v(x)$ are differentiable functions of x whose values lie in $[a, b]$, then

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt = f(v(x))\frac{dv}{dx} - f(u(x))\frac{du}{dx}.$$

To prove the rule, let F be an antiderivative of f on $[a, b]$. Then

$$\int_{u(x)}^{v(x)} f(t)\,dt = F(v(x)) - F(u(x)).$$

Differentiating both sides of this equation with respect to x gives the equation we want:

$$\begin{aligned}\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt &= \frac{d}{dx}[F(v(x)) - F(u(x))] \\ &= F'(v(x))\frac{dv}{dx} - F'(u(x))\frac{du}{dx} \qquad \text{Chain Rule} \\ &= f(v(x))\frac{dv}{dx} - f(u(x))\frac{du}{dx}.\end{aligned}$$

Use Leibniz's Rule to find the derivatives of the functions in Exercises 27–29.

28. $f(x) = \displaystyle\int_{\cos x}^{\sin x} \frac{1}{1 - t^2}\,dt$

Exercises 8.2

Using Substitution

Evaluate the integrals in Exercise 25–30 by using a substitution prior to integration by parts.

26. $\displaystyle\int_0^1 x\sqrt{1-x}\,dx$

Evaluating Integrals

Evaluate the integrals in Exercises 31–52. Some integrals do not require integration by parts.

52. $\displaystyle\int x^2 \tan^{-1}\frac{x}{2}\,dx$

Reduction Formulas

In Exercises 63–67, use integration by parts to establish the reduction formula.

63. $\displaystyle\int x^n \cos x\,dx = x^n \sin x - n\int x^{n-1}\sin x\,dx$

64. $\displaystyle\int x^n \sin x\,dx = -x^n \cos x + n\int x^{n-1} \cos x\,dx$

68. Use Example 5 to show that

$$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \begin{cases} \left(\dfrac{\pi}{2}\right)\dfrac{1\cdot 3\cdot 5\cdots(n-1)}{2\cdot 4\cdot 6\cdots n}, & n \text{ even} \\ \dfrac{2\cdot 4\cdot 6\cdots(n-1)}{1\cdot 3\cdot 5\cdots n}, & n \text{ odd} \end{cases}$$

69. Show that

$$\int_a^b \left(\int_x^b f(t)\, dt \right) dx = \int_a^b (x - a) f(x)\, dx.$$

(In Ex.69, let f(x) be continuous.)

70. Use integration by parts to obtain the formula

$$\int \sqrt{1 - x^2}\, dx = \frac{1}{2} x \sqrt{1 - x^2} + \frac{1}{2} \int \frac{1}{\sqrt{1 - x^2}}\, dx.$$